MATH0120 Complex Function Theory

Year:	2026–2027
Code:	MATH0120
Level:	7 (UG), 7 (PG)
Normal student group(s):	UG: Y4 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	80% examination, 20% coursework
Normal Pre-requisites:	MATH0013 (Analysis 3: Complex Analysis)
Lecturer:	Prof Y Petridis

Course Description and Objectives

Complex function theory has outmost beauty and is of paramount importance in mathematics, pure and applied. It is of essential use in the theory of modular functions, elliptic curves over $\mathbb{C}$, and analytic number theory: analytic continuation of the Riemann zeta function, factorisation in terms of its zeros (and similar results for other L-functions). It is the basis for Riemann surfaces as complex one-dimensional manifolds and their uniformisation, and as precursor to Complex and Kähler manifolds. It is linked to conformal mapping as applied e.g. to fluid mechanics.

In this module, we will expand on the notions encountered in MATH0013 and study far-reaching theorems in this field and some of their applications.

Recommended Texts

- L Ahlfors, *Complex Analysis: An Introduction to the Theory of Analytic Functions of One Complex Variable* 3rd ed (AMS)
- EM Stein and R Shakarchi, *Complex Analysis* (Princeton)
- JB Conway, *Functions of one complex variable I* 2nd ed (Springer)

Detailed Syllabus

- Review of complex analysis. The Argument principle and Rouché's theorem. The open mapping theorem, and the maximum principle. Hadamard's three circles theorem and Phragmén Lindelöf theorem.
- Harmonic functions, Schwarz reflection principle, Poisson kernel. Series of analytic functions and the Mittag-Leffler theorem.
- The Riemann mapping theorem. Explicit conformal mappings and the Schwarz–Christoffel formula. The Dirichlet problem and its solution: subharmonic functions, Harnack's inequality.
- Infinite products, and Weierstrass factorisation. Jensen's formula. The gamma function and Stirling asymptotics.